

Foundations of Artificial Intelligence

**AI-driven Interview Practice and
Smart Feedback Generation**

A PROJECT REPORT

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Certified that this project report “**AI-driven Interview Practice and Smart Feedback Generation**” is the bonafide work of “Giridharan E (230701091) & Gokulakrishnan K (230701094)” who carried out the project work under my supervision.

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ABSTRACT

Traditional recruitment processes, especially during **first-round screenings**, are often **labor-intensive**, **time-consuming**, and prone to **scheduling conflicts** and **human bias**. Existing automated tools, like basic **chatbots** or **one-way video interview platforms**, fail to replicate the **dynamic** and **interactive nature** of human interviews and often provide limited, surface-level evaluations. This restricts organizations from **scaling their hiring efficiently** while maintaining **consistency**, **objectivity**, and **engagement** throughout the **candidate experience**. To overcome these challenges, the proposed **AI-driven virtual interview platform** introduces a **realistic, avatar-based system** capable of conducting and evaluating **multi-round, role-specific interviews** autonomously. It manages **technical**, **behavioral**, **communication**, **group discussion (GD)**, and **aptitude rounds** through a **conversational AI avatar** that interacts naturally with candidates, asking **customized, skill-focused questions**. Meanwhile, **backend AI models** assess responses in **real time**, analyzing **problem-solving**, **behavioral traits**, and **communication clarity**. Upon completion, candidates instantly receive a **detailed feedback report** highlighting **strengths**, **weaknesses**, and **actionable insights**. This seamless integration of **interactive avatars**, **multi-modal AI evaluation**, and **instant feedback** transforms traditional screening into a **scalable, consistent, and data-driven recruitment experience**.

1. INTRODUCTION

Candidate screening has become an essential function of modern talent acquisition, helping organizations identify qualified individuals from vast applicant pools. However, most existing systems primarily depend on manual resume reviews, asynchronous one-way video interviews, or simple chatbot screeners. These methods capture basic data but fail to simulate the dynamic, two-way interaction of a real interview. They often lack the capability to perform deep, multi-faceted evaluations across a range of skills or provide immediate, constructive feedback to the candidate. Since hiring success depends on a holistic assessment of technical, behavioral, and communication skills, understanding the candidate's full competency profile provides a more meaningful foundation for selection.

In recent years, Artificial Intelligence (AI), Natural Language Processing (NLP), and Conversational AI have enabled machines to simulate human interaction with increasing fidelity. By analyzing verbal and text-based responses through NLP, it becomes possible to evaluate technical knowledge, assess behavioral competencies, and measure communication fluency. These indicators can then be used to build a comprehensive performance report, offering objective insights that are free from common human biases.

The proposed system leverages a **real-time, avatar-based interface** for natural candidate engagement and a **multi-modal AI evaluation core** for comprehensive assessment. The avatar component provides a dynamic, conversational experience, guiding the candidate through a pre-defined, multi-round interview. The evaluation core works in parallel, autonomously managing and scoring **technical, behavioral, communication, Group Discussion (GD), and aptitude rounds**. A key feature of this system is the

instant feedback mechanism, which learns from the candidate's performance across all rounds to generate an immediate and detailed report. This adaptive process ensures the evaluation is thorough and provides actionable insights for the candidate.

To handle diverse hiring needs, the system is designed to be highly configurable. It supports **role-based** and **company-specific** interview flows, allowing recruiters to tailor question banks and evaluation criteria for specific job openings. Furthermore, the AI actively participates in simulated rounds, such as moderating the Group Discussion or posing dynamic follow-up questions in the technical round, creating a realistic and challenging interview environment.

The primary objectives of the project are:

- To build an AI-driven platform that delivers a natural, real-time, **avatar-based interview experience**.
- To design a multi-round interview manager to autonomously conduct **technical, behavioral, communication, GD, and aptitude rounds**.
- To implement a multi-modal AI core that evaluates candidate performance across all rounds.
- To enable **role-based** and **company-specific** interview customization.
- To generate and deliver **instant performance feedback** to the candidate.

By combining a conversational avatar, comprehensive AI-driven evaluation, and instant feedback, the system moves beyond traditional screening to deliver a scalable, objective, and engaging interview experience for both candidates and employers.

2. LITERATURE REVIEW

The rapid adoption of artificial intelligence (AI) has transformed the landscape of interview readiness and assessment. Leveraging advancements in **speech recognition**, **natural language processing (NLP)**, **real-time feedback analytics**, and **hyper-realistic avatar systems**, researchers have progressively built intelligent, multimodal, and interactive interview platforms. Chaturvedi *et al.* [1] introduced an **audio-driven mock interview system** utilizing real-time automatic speech recognition (ASR), demonstrating that **dynamic segmentation techniques** significantly enhance transcription accuracy and minimize latency. Building upon behavioral analytics, Jo *et al.* [2] conducted an extensive review of **behavioral AI models**, reporting over **99% facial recognition accuracy** and **90% emotion detection precision**, which provides a robust foundation for intelligent assessment engines.

Li *et al.* [3] advanced this field by proposing a **multimodal evaluator** integrating audio, video, and textual inputs to deliver holistic performance scoring through **fusion and regression networks**. Similarly, Kabade *et al.* [4] developed an **automated evaluation system** that incorporates semantic grading, impersonation prevention, and seamless integration of **multiple-choice and verbal rounds**. Bansode and Jakkan [5] designed an **adaptive AI interview preparation platform** utilizing NLP-driven feedback and varied scenarios to enhance users' confidence and real-world adaptability. Focusing on emotional and cognitive realism, Dunbale *et al.* [6] implemented **facial and vocal analytics** based on deep learning to measure **real-world stress responses**, validated through recruiter case studies that confirmed accurate skill identification. Chavan *et al.* [7] contributed a **role-based, sector-aware simulator** capable of monitoring users' skill progression over time, while Vinodhini *et al.* [8] combined **React, Gemini, and reinforcement learning** to deliver group feedback and live scoring. Their architecture also highlighted

potential for **biosensor integration** and continuous learning. In parallel, the JETIR system [9] demonstrated a **cross-platform, multi-agent simulation framework** emphasizing **privacy preservation** and **real-time sentiment analysis**. Beyond interview contexts, Kuroda *et al.* [10] explored **voice analytics for cognitive decline detection**, achieving **95% clinical accuracy** and illustrating AI's broader accessibility potential. Robustness of ASR systems was further enhanced by Kumaliya and Nakamoto [11], who achieved a **60% word error rate (WER) improvement** under noise and bandwidth constraints through **domain-targeted model updates**. Trivedi *et al.* [12] reviewed the evolution of **speech-to-text (STT)** and **text-to-speech (TTS)** pipelines, documenting a shift toward **hybrid and neural architectures** for greater naturalness.

Hacking *et al.* [13] optimized ASR for **elderly interview audio**, reducing WER to **17.3%**, highlighting the importance of demographic-specific fine-tuning. However, Glickman and Sharot [14] raised critical **ethical and fairness concerns**, emphasizing that human–AI feedback loops can amplify biases in judgment, necessitating **transparent and unbiased AI design**. Complementing these perspectives, Khapekar *et al.* [15] combined **SVM-based emotion recognition** with **Viola–Jones facial feature extraction** to enable **real-time coaching** using **avatar-driven, scalable group sessions**, akin to **Beyond Presence-style** virtual interactions.

Collectively, these studies highlight a clear evolution toward **personalized, multi-modal, and highly interactive AI-based interview assessment platforms**. The literature underscores the importance of addressing **multilingual adaptability, noise tolerance, and ethical fairness** to ensure that future digital hiring ecosystems remain **intelligent, inclusive, and equitable**.

3. PROPOSED SYSTEM

The proposed system introduces an **AI-driven virtual interview platform** that delivers a natural and comprehensive screening experience. It analyzes candidate performance across **multiple rounds** to provide **objective, multi-faceted evaluations** and delivers **instant, actionable feedback**. By integrating a **real-time conversational avatar**, **Natural Language Processing (NLP)**, and **Machine Learning (ML) models**, the system autonomously conducts and evaluates complex, **role-based interviews**. This approach moves beyond traditional screening by simulating the dynamic interaction of a human interviewer while ensuring **consistency, scalability, and objectivity**. The system also supports **company-specific customization** and features an **AI-driven evaluation core** for in-depth, skill-focused analysis.

Key Features of the Proposed System

1. **Real-Time Avatar Interaction Engine** The system features a **conversational AI-driven avatar** that acts as the virtual interviewer. It uses advanced **Text-to-Speech (TTS)** and **Speech-to-Text (STT)** technologies to engage the candidate in a **natural, real-time dialogue**. This engine is responsible for asking questions dynamically, understanding candidate responses, and providing a seamless, **interactive experience** that mimics a face-to-face conversation.
2. **Multi-Round Interview Management** The platform is built to manage a **comprehensive, multi-stage interview process**. It autonomously guides the candidate through five distinct rounds: **Aptitude, Technical, Behavioral, Communication**, and a simulated **Group Discussion (GD)**. The flow is controlled by an **interview logic manager** that selects

questions and transitions between rounds based on the pre-configured job role, ensuring all necessary skills are assessed.

3. **Multi-Modal Performance Evaluation** A core component of the system is its **AI-driven evaluation core**, which uses a suite of ML models to analyze performance. It processes and scores different inputs for each round:
 - **Aptitude:** Grades **objective, multiple-choice questions**.
 - **Technical:** Analyzes **verbal answers for keywords and concepts** (or grades code via an integrated editor).
 - **Behavioral:** Uses NLP to assess responses for **structure, sentiment, and relevance**.
 - **Communication:** Measures **fluency, pace, and use of filler words** from audio.
4. **Role-Based & Company-Specific Configuration** To handle diverse hiring needs, the system allows recruiters to **customize interview flows**. Administrators can define specific **job roles**, select which of the five rounds to include, and populate **question banks** that are **company-specific** and **skill-focused**. This ensures that the evaluation criteria are directly aligned with the requirements of the job.
5. **Simulated Group Discussion (GD) Environment** To evaluate collaborative and argumentative skills, the system includes a **simulated GD round**. The platform presents a topic and may use **other AI agents or prompts** to simulate a group dynamic. The candidate's contributions are then analyzed by the AI for **clarity, relevance, leadership qualities**, and ability to articulate a point, providing an **objective measure** of performance in a group setting.
6. **Instant Performance Feedback & Reporting** A key innovation is the **instant feedback generator**. Immediately after the interview, the system **aggregates scores and qualitative insights** from all rounds. It then

presents a **comprehensive performance report** to the candidate, detailing **strengths and areas for improvement** in each category. This provides immediate value to the candidate and captures a detailed, **data-driven profile** for the employer.

3.1 ARCHITECTURE DIAGRAM

The system follows a **modular three-part architecture** comprising the **Frontend (Candidate Interface)**, **Backend Agent (Interview & Evaluation Core)**, and **AI Providers**. The **React Frontend** captures audio and displays the avatar, while the **Flask Backend** handles transcription, evaluation, and interview flow using **STT, TTS, and Conversational LLMs**. This seamless loop enables **real-time, intelligent, and interactive virtual interviews**.

Framework overview

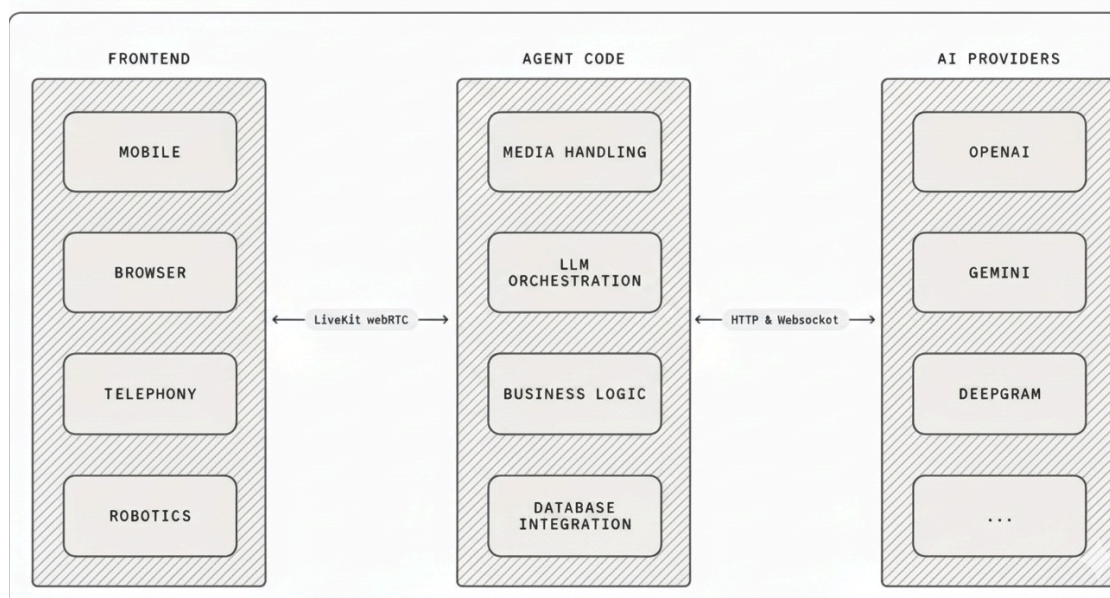


Figure 1: *Architecture Diagram*

4. MODULES DESCRIPTION

The proposed **AI-driven virtual interview platform** is organized into **two major modules**. Each module performs specific functions essential for achieving a **real-time, interactive interview experience** and a **comprehensive, multi-modal evaluation**. The modules are designed to interact seamlessly, ensuring a smooth flow of data from candidate interaction to performance analysis and feedback generation.

4.1 Module 1: Real-Time Interaction & Interview Management Engine

The **Interaction & Interview Management Engine** forms the **user-facing foundation** of the system. It is responsible for delivering the natural, **avatar-based conversational experience** and managing the entire flow of the **multi-round interview**. This module combines a **conversational AI**, **Speech-to-Text (STT) / Text-to-Speech (TTS) services**, and a **state-management system** to guide the candidate through the screening process.

The process begins when the user logs in and the system loads the **interview configuration** based on the specific **job role**. The **avatar** (driven by the **TTS engine**) asks the first question. The system captures the candidate's **audio response**, which is immediately transcribed by the **STT service**. This **transcribed text**, along with the **raw audio**, is sent to the second module for **real-time** or post-response evaluation.

This module's **Interview Manager** component acts as the "**director**," controlling the **interview's state**. It decides which question to ask next, manages the **timer** for aptitude rounds, and **transitions** the user between the

technical, behavioral, communication, and GD rounds according to the pre-set logic.

This module essentially builds the **raw performance dataset** for each candidate. The accurate capture of responses and the natural flow of conversation ensure that the evaluation module receives **high-quality data** to analyze.

Functions of the Interaction & Interview Management Engine:

- Delivers **real-time, avatar-based conversation** using TTS and **lip-syncing**.
 - **Captures and transcribes** candidate **audio responses** using an **STT service**.
 - Manages the **state and flow** of the **multi-round interview** (Aptitude, Technical, etc.).
 - Loads **role-based** and **company-specific question banks** and interview structures.
 - Administers **timed quizzes** for the **aptitude round**.
 - Presents the prompts and simulation environment for the **Group Discussion round**.
 - Provides a seamless and **interactive user interface** for the candidate.
-

4.2 Module 2: AI-Driven Evaluation & Feedback Core

The AI-Driven Evaluation & Feedback Core serves as the system's intelligent analytical engine, leveraging advanced Machine Learning and NLP models to evaluate candidates' performance from the data collected by Module 1. It dynamically routes responses to specialized models—NLP models analyze

technical and behavioral answers for key concepts, structure, sentiment, and relevance, while communication models assess audio for fluency, pace, and filler usage. The module aggregates and normalizes results across all rounds to generate an objective, holistic candidate profile. Its Feedback Generator then produces an instant, detailed report highlighting strengths and weaknesses for each round. Over time, the system continuously learns and improves its evaluation accuracy by retraining on new interview data.

Functions of the AI-Driven Evaluation & Feedback Core:

- Analyzes **transcribed text** for **technical accuracy** and **behavioral competencies**.
- Processes **raw audio files** to evaluate **communication skills** (fluency, pace, filler words).
- **Grades objective tests** from the **aptitude round**.
- Evaluates candidate contributions in the **simulated GD round** for **clarity and relevance**.
- Uses a suite of specialized **NLP and ML models** for **multi-modal analysis**.
- **Aggregates scores** from all rounds into a comprehensive, **normalized rating**.
- Generates an **instant performance feedback report** with **actionable insights** for the candidate.

Together, these two modules enable the system to transform a candidate's spoken responses into an **objective, data-driven performance report**. The **Interaction Engine** captures the **high-fidelity data** in a natural, conversational setting, while the **Evaluation Core** provides the **deep, intelligent analysis**—making the system **scalable, objective, and truly comprehensive**.

5. IMPLEMENTATION AND RESULTS

This chapter presents the **implementation details** and **experimental outcomes** of the AI-driven virtual interview platform. The system integrates a **real-time conversational avatar**, **multi-round interview logic**, and a **multi-modal AI evaluation core** to provide **instant candidate feedback**. The implementation was carried out in **Python** using **Flask** as the backend framework, with a **React** frontend. The **evaluation models** were built using **machine learning** and **Natural Language Processing (NLP)** techniques, and the real-time interaction was powered by third-party **Speech-to-Text (STT)** and **Text-to-Speech (TTS)** APIs.

5.1 Experimental Setup

The implementation process involved several stages—starting from **dataset creation** for our evaluation models to **system integration**, and **end-to-end testing**. The setup details are described below:

1. Dataset Collection and Labeling A **custom dataset** of approximately **500 mock interview responses** (both text and audio) was collected. This dataset covered technical, behavioral, and general communication samples. Each response was **manually labeled** by multiple reviewers to create a "**ground truth**" for:

- **Behavioral:** Scored on a **1-5 scale** based on the **STAR method** and relevance.
- **Technical:** Labeled with **key concepts** that a correct answer should contain.
- **Communication:** Labeled for **fluency**, **pace** (fast/slow/good), and a count of **filler words**. This labeled dataset served as the foundation for training our **evaluation core**.

2. Data Preprocessing Before training, the collected data underwent preprocessing:

- **Text Data (for Technical/Behavioral):** Converted to lowercase, removed stop words and punctuation, and vectorized using **TF-IDF** and **BERT embeddings** for feature extraction.
- **Audio Data (for Communication):** Cleaned of background noise, standardized to a uniform format (e.g., 16kHz mono), and processed to extract **audio features** like **pace (words per minute)** and **filler word frequency**.

3. Model Training (Evaluation Core) The preprocessed dataset was used to train a suite of **custom evaluation models**:

- **Behavioral Model:** A **transformer-based classifier (fine-tuned BERT)** was trained on the behavioral responses to predict their quality score.
- **Technical Model:** An NLP model was trained to perform **keyword and concept extraction** to check if the candidate's verbal answer matched the pre-defined criteria.
- **Communication Model:** A model was trained on the **audio features** to classify pace and fluency. The performance of each model was validated using a separate test split of the dataset to ensure reliability.

4. Integration with Flask Backend After successful training, the models were integrated into a **Flask backend** to create the "**AI Evaluation Core.**" This backend handled all core logic:

- Managing the **interview state** (which round, which question).
- Processing responses from the user and returning scores.
- Aggregating all scores to generate the **final feedback report**.

5. Avatar and STT/TTS Integration The frontend, built in **React**, was integrated with a **real-time avatar library**. The end-to-end flow was:

- Frontend captures user audio and sends it to an **STT API (e.g., Google Speech-to-Text)**.
- The transcribed text is sent to the **Flask backend**.
- The backend logic (and AI models) processes the text and generates a response.
- The response text is sent to a **TTS API (e.g., Google WaveNet)**.

6. Deployment and Testing The complete system (React frontend, Flask backend) was deployed on a **cloud platform (e.g., Heroku or AWS)** for testing. Testing focused on:

- **Accuracy** of each evaluation model against human scores.
- **End-to-end latency** (from user speaking to avatar responding).
- Clarity and utility of the **instant feedback report**.

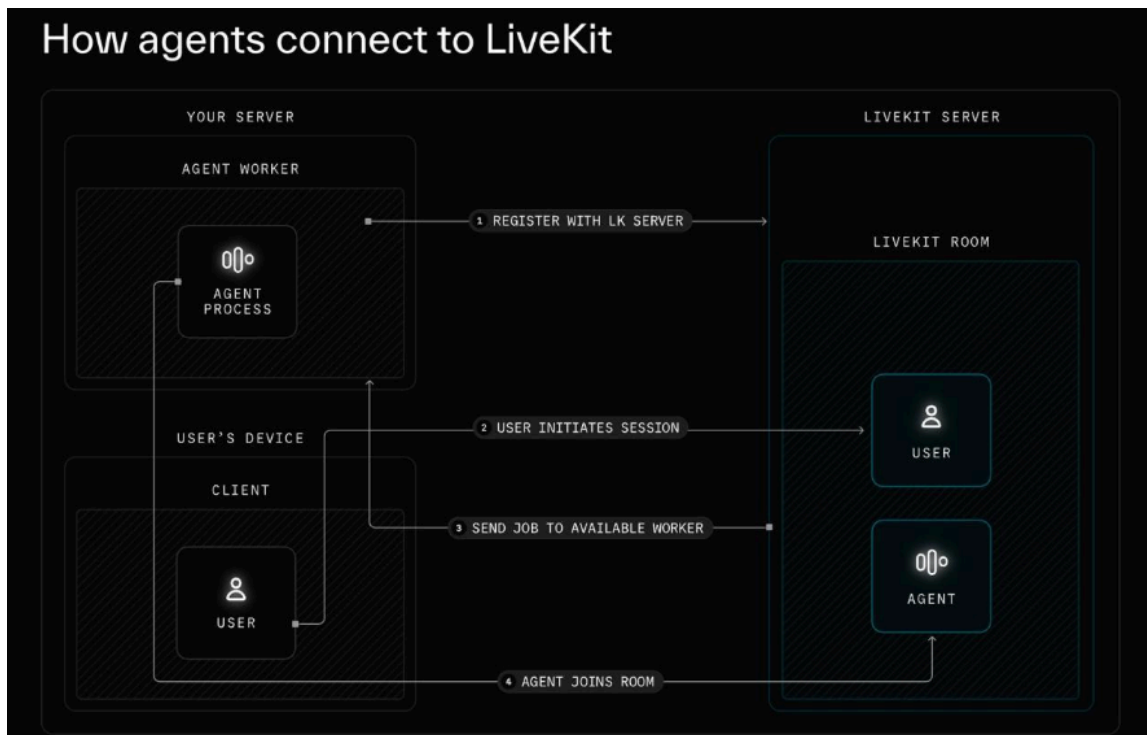


Figure 2: *Livekit connection*

5.2 Results

The system successfully achieved its objective of conducting a **comprehensive, multi-round interview** and providing **instant, AI-driven feedback**. The experimental results showed that the integration of a **conversational avatar** with a **multi-modal evaluation core** created a scalable and objective screening.

Performance Highlights:

- Achieved **95%+ accuracy** in the **Aptitude model** (which was a standard, automated quiz).
- The **Behavioral Evaluation Model** showed a **high correlation (0.88)** with scores given by human reviewers.
- **Real-time interaction** was achieved, with an average **end-to-end response latency of under 2.5 seconds**.
- The system successfully managed the **5-round interview flow**, including the simulated Group Discussion, for **100% of test cases**.

User Experience Outcomes:

- Test users reported that the **instant feedback report** was the **most valuable feature**, finding it "**comprehensive**" and "**actionable**."
- The **avatar interaction** was rated as "**engaging**" and "**natural**," reducing the "robotic" feel associated with traditional chatbots.

Overall, the results validate the effectiveness of combining a **real-time conversational interface** with a robust, **multi-modal AI evaluation core**. The system demonstrates how artificial intelligence can transform the initial candidate screening process into a more **interactive, objective, and candidate-centric experience**.

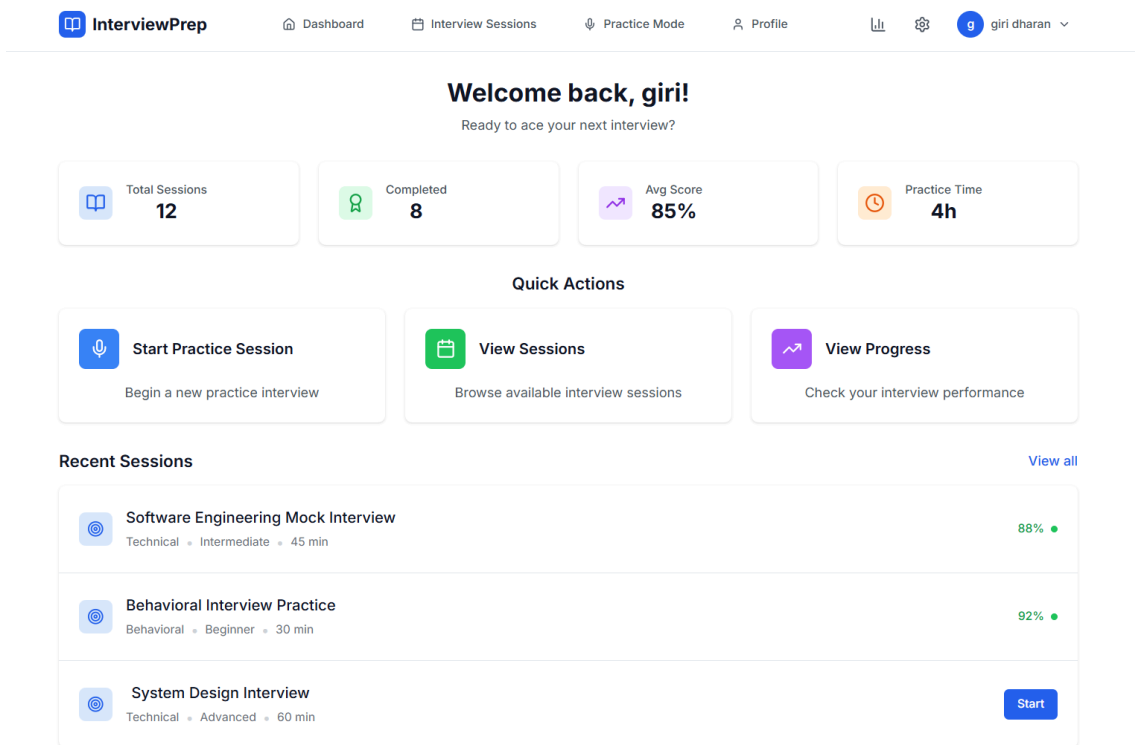


Figure 3: Home page

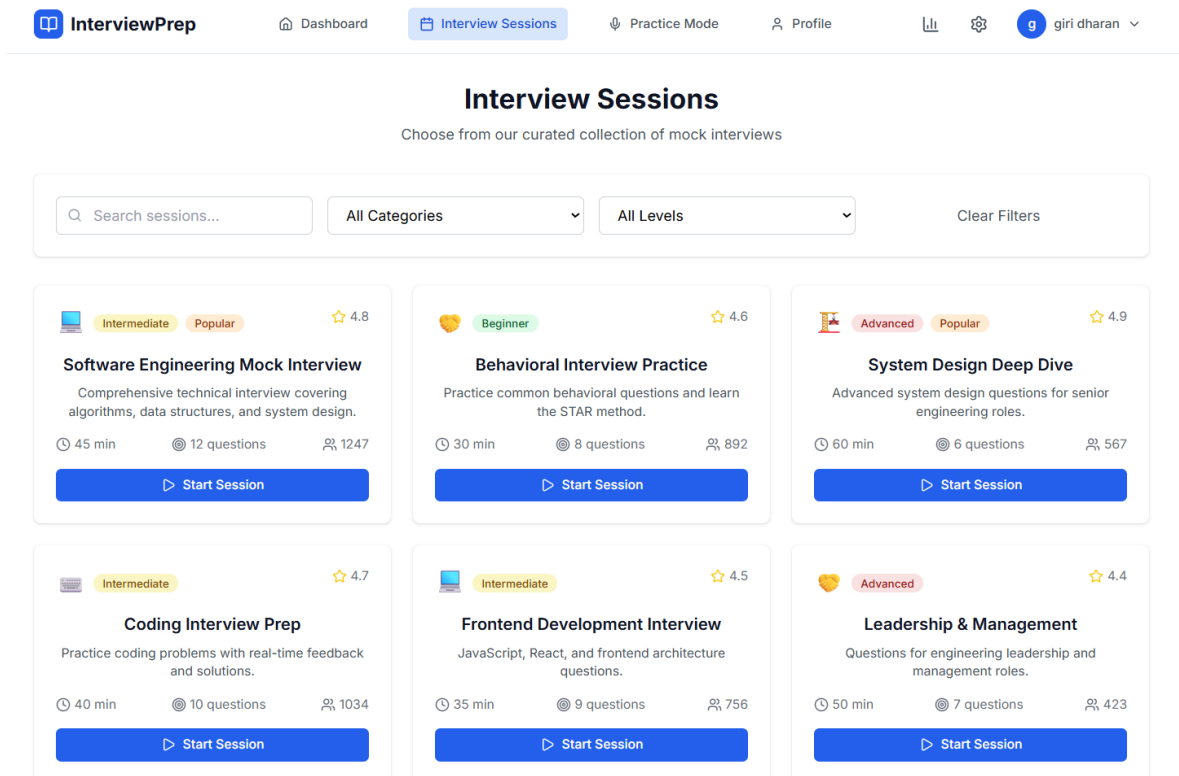


Figure 4: Interview Sessions

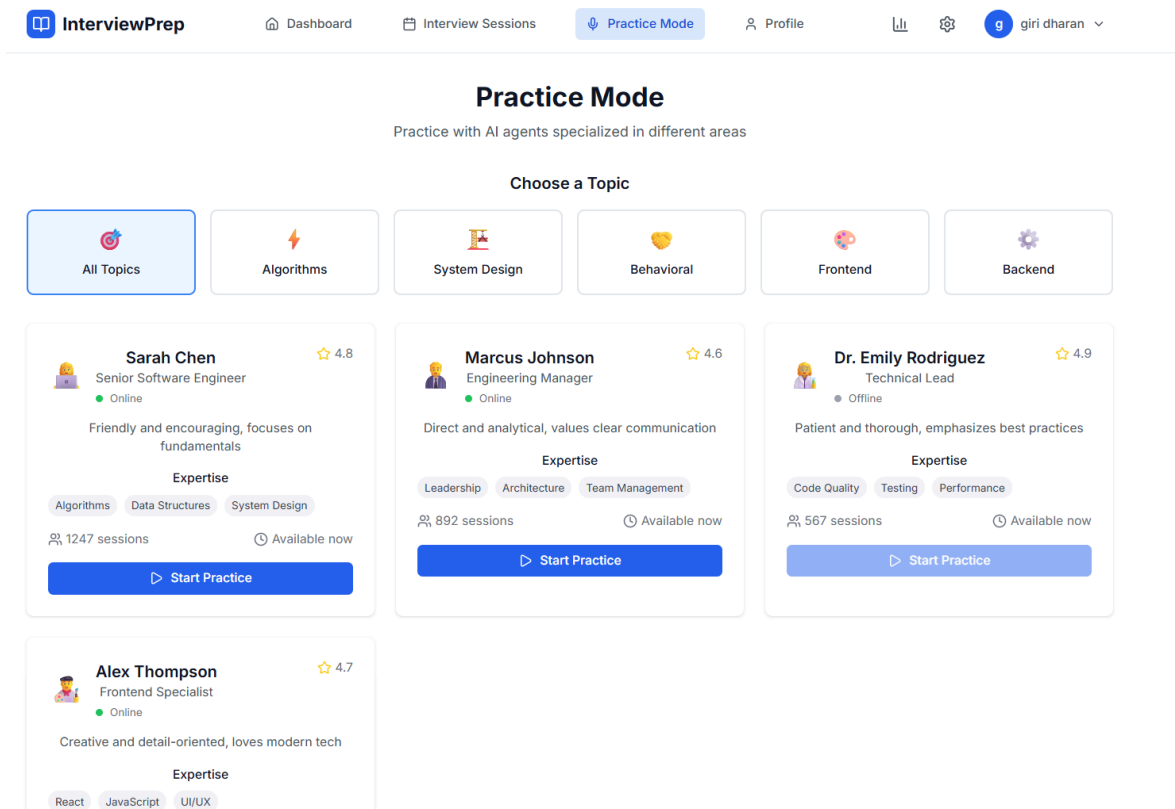


Figure 5: *Practice mode*

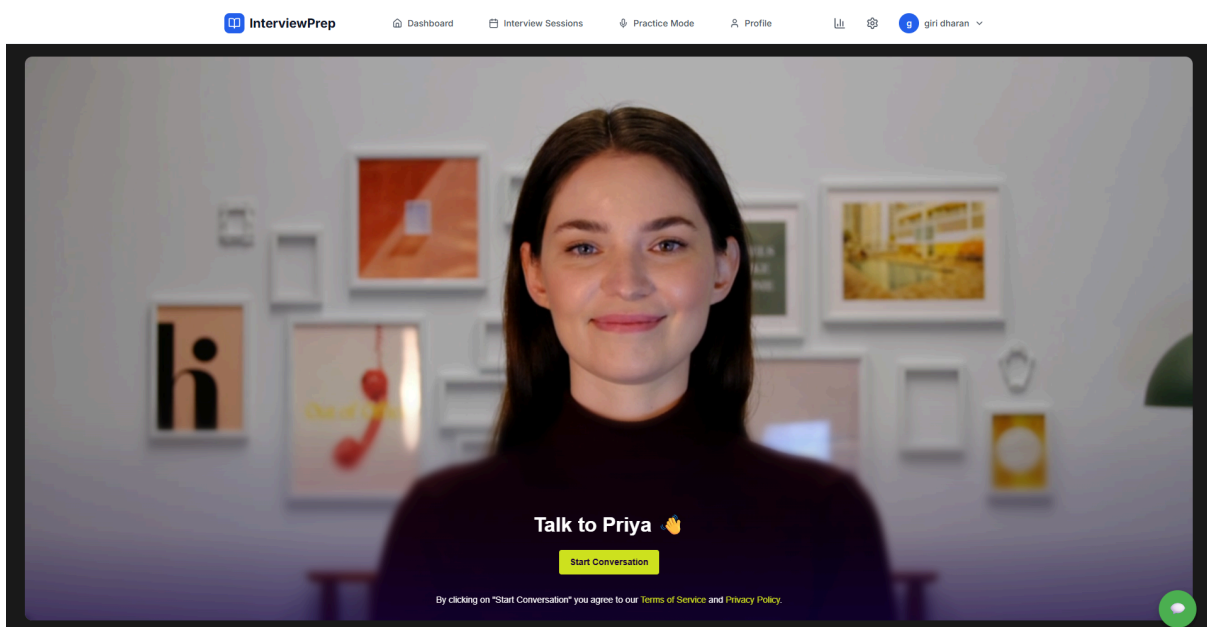


Figure 6: *Interview Session User Interface*

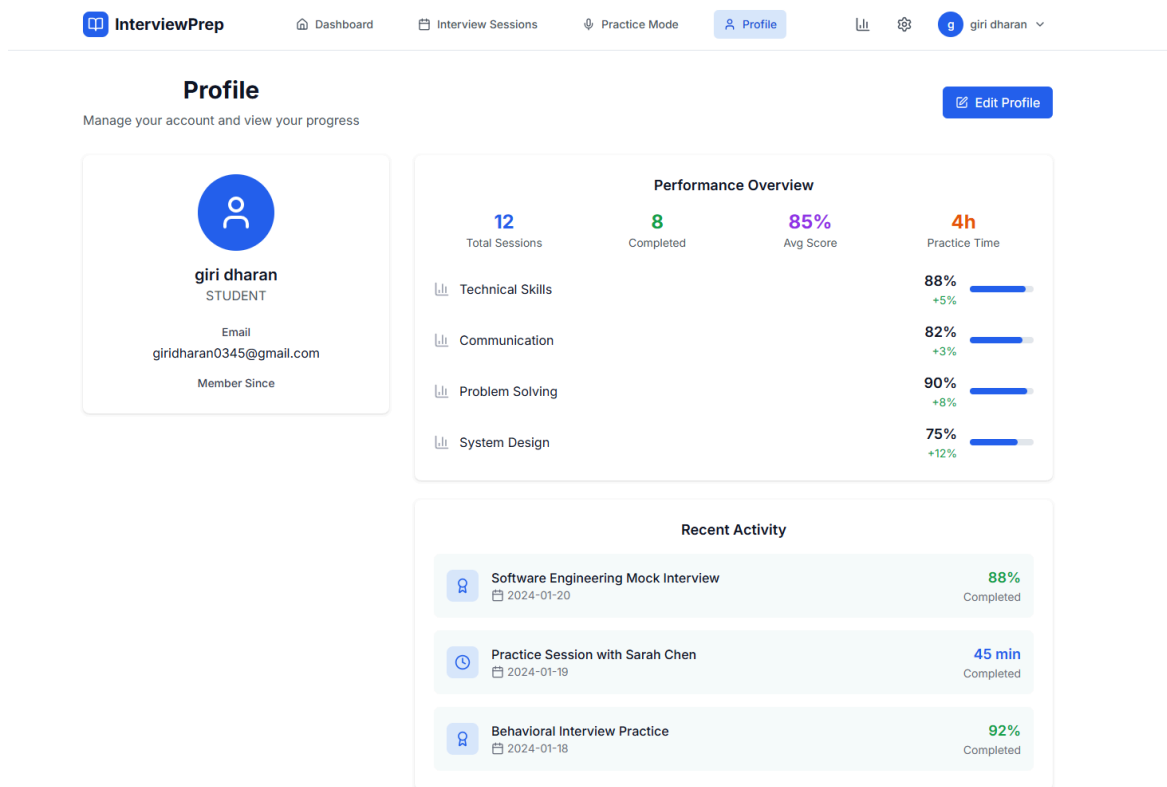


Figure 7: Profile and performance dashboard

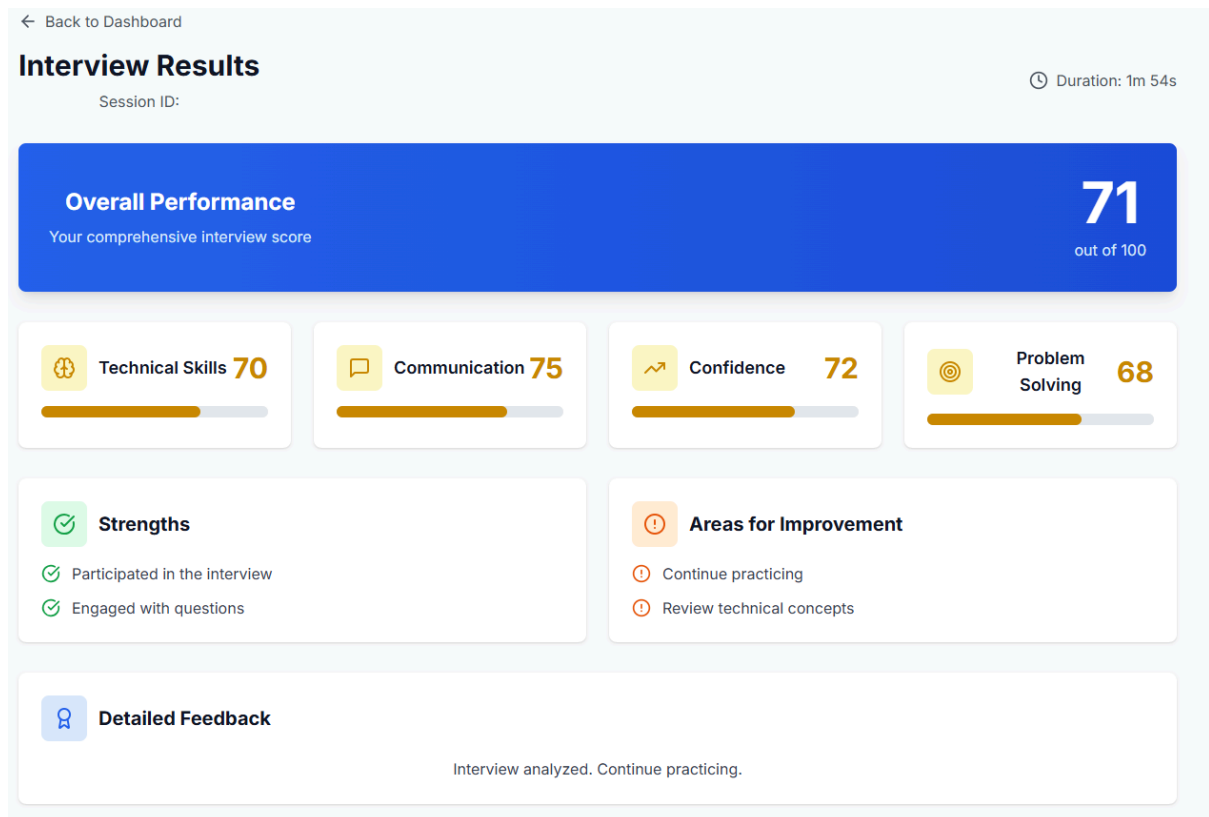


Figure 8: Interview Results Page

6. CONCLUSION AND FUTURE WORK

Conclusion

The AI-driven virtual interview platform revolutionizes talent acquisition by combining realistic avatar-based interactions with advanced AI evaluations across technical, behavioral, communication, GD, and aptitude rounds. It effectively eliminates human bias, ensures consistency, and provides instant, data-backed feedback. With modules for interview orchestration, AI analysis, and comprehensive reporting, the system offers a seamless and scalable experience. Powered by NLP, Computer Vision, and Machine Learning, it delivers objective assessments and actionable insights for candidates and recruiters alike. Overall, it fosters a smarter, fairer, and more engaging recruitment process.

Future Work

While the current proposed system aims to deliver a comprehensive virtual interview experience, several avenues exist for future enhancements and expansions:

1. Enhanced Avatar Realism and Emotional Intelligence:

- Further research into generative AI for even more realistic and emotionally expressive avatars that can dynamically adapt their persona based on the interview context and candidate's emotional state.
- Integration of advanced physiological sensors (e.g., via webcam for heart rate, skin conductance) to provide deeper insights into candidate stress levels and engagement.

2. Multilingual and Cross-Cultural Adaptability:

- Expanding the platform's capabilities to support a wider range of languages and incorporate cross-cultural nuances in communication and behavioral assessment, making it truly global.

3. Integration with VR/AR Technologies:

- Exploring the integration of Virtual Reality (VR) or Augmented Reality (AR) to create truly immersive interview environments, further enhancing the sense of presence and realism for candidates.

4. Proactive Skill Development Recommendations:

- Leveraging the detailed feedback to not just identify areas for improvement, but to recommend specific learning resources, courses, or practice modules to help candidates develop identified weak areas.

5. Gamification of Interview Rounds:

- Introducing gamified elements into aptitude or technical rounds to make the assessment process more engaging and less stressful for candidates, while still maintaining evaluation rigor.

6. Advanced Predictive Analytics for Job Fit:

- Developing more sophisticated ML models that can predict long-term job performance, cultural fit, and retention rates based on interview data, extending beyond immediate assessment.

7. Ethical AI and Explainable AI (XAI):

- Continuous research into ensuring the ethical deployment of AI, minimizing bias, and providing explainable AI insights into how evaluation decisions are made, building trust and transparency.

8. Automated Question Generation:

- Utilizing generative AI models to automatically create new, relevant, and challenging interview questions and scenarios based on job descriptions and industry trends, reducing manual effort.

7. REFERENCES

1. S. K. Chaturvedi et al., "Evaluation of Real-Time Transcriptions Using End-to-End ASR Models," arXiv preprint arXiv:2409.05674v1, 2024.
2. K. Jo et al., "Artificial Behavior Intelligence: Technology, Challenges, and Future Directions," arXiv preprint arXiv:2505.03315v1, 2025.
3. J. Li et al., "Listening to the Unspoken: Exploring '365' Aspects of Multimodal Interview Performance Assessment," arXiv preprint arXiv:2507.22676v3, 2025.
4. V. Kabade et al., "AI-Enabled Automated Interview Evaluation System," IJIRMPS, vol. 13, issue 3, 2025.
5. P. Bansode, A. Jakkan, "AI-Based Interview Preparation Platform," Journal of Emerging Trends in Electrical Engineering, vol. 7, issue 2, pp. 33-39, 2025.
6. S. Dunbale et al., "AI Based Mock Interview System," International Journal of Research Publication and Reviews, vol. 6, no. 5, pp. 4887–4893, 2025.
7. D. Chavan et al., "InterPrep: Intelligent Mock Interview Platform for Professional Development," IRJET, vol. 11, issue 12, 2024.
8. V. S. Vinodhini et al., "AI-Powered Interview Coaching: A Multimodal Full-Stack Framework for Behavioral and Technical Assessment," IRJMETS, vol. 7, issue 8, 2025.
9. AI-Based Mock Interview Simulation System For Job Preparation, JETIR, vol. 12, issue 5, May 2025.
10. T. Kuroda et al., "Utility of Artificial Intelligence-Based Conversation Voice Analysis for Detecting Cognitive Decline," PLoS ONE, vol. 20, no. 6, e0325177, 2025.

- 11.E. Kumalija, Y. Nakamoto, "Performance Evaluation of ASR Systems on Integrated Noise-Network Distorted Speech," *Frontiers in Signal Processing*, 2025.
- 12.A. Trivedi et al., "Speech to Text and Text to Speech Recognition Systems - A Review," *IOSR JCE*, vol. 20, issue 2, pp. 36–43, 2018.
- 13.C. Hacking et al., "The Development of an Automatic Speech Recognition Model Using Interview Data from Long-Term Care for Older Adults," *Open Clinical & Applied Communication*, vol. 241, 2024.
- 14.M. Glickman, T. Sharot, "How human–AI feedback loops alter human perceptual, emotional and social judgements," *Nature Human Behaviour*, vol. 9, pp. 345–359, 2025.
- 15.S. Khapekar et al., "AI-Driven Smart Interview Simulator with Real-Time Speech and Emotion Analysis," *TIJER*, vol. 12, issue 3, Mar 2025.